A counterexample family to Simon's Le Roy-type criterion

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Abstract

We disprove the proposed necessity of the logarithmic infinite-divisibility criterion for complete monotonicity of a Le Roy-type Mittag–Leffler function. With $\alpha = 1/3$ and $\gamma = 2$, a nonempty open interval of parameters β below the proposed threshold still gives completely monotone functions. The proof constructs a signed density by beta integrals, proves strict positivity at the threshold, and shows that positivity persists across that threshold. The existence of the interval is proved analytically; its width is not assigned a numerical value.

Task 1: Resolve MathDB problem 349140

Write

$$F_{\alpha,\beta}^{(\gamma)}(z) = \sum_{n=0}^{\infty} \frac{z^n}{\Gamma(\beta+\alpha n)^\gamma}, \qquad \alpha, \beta, \gamma > 0.$$

The conjecture in Section 3 of [1] asserts the equivalence

$$F_{\alpha,\beta}^{(\gamma)}(-x) \text{ is completely monotone for } x > 0 \quad \Longleftrightarrow \quad z + \gamma(z^{\beta-\alpha} - z^\beta) \le 1 \quad (0 < z < 1). \tag{1}$$

Its sufficient direction was already proved there. We refute necessity.

Theorem 1. *Define*

$$\beta_c = \max_{0<y<1} \frac{1}{3}\left(1 + \frac{\log((1+y+y^2)/2)}{\log y}\right). \tag{2}$$

Then $1/3 < \beta_c < 1/2$, and there exists $\varepsilon > 0$ such that, for every $\beta \in (\beta_c - \varepsilon, \beta_c)$, $F_{1/3,\beta}^{(2)}(-x)$ is strictly completely monotone on $(0,\infty)$, whereas the right side of (1) *fails.*

The proof below defines a permissible positive ε in terms of convergent integrals and extrema on compact sets. It does not claim that any particular decimal or rational β has been certified.

The threshold

Set $a = 1/3$ and $b = 2/3$ throughout. With $z = y^3$, the proposed condition is equivalent to

$$1 + y + y^2 \ge 2y^{3\beta-1} \quad (0 < y < 1), \tag{3}$$

because the difference between its two sides in (1) is $(1-y)(1+y+y^2-2y^{3\beta-1})$. Taking logarithms shows that (3) holds exactly when $\beta \ge \beta_c$. The function being maximized tends to a at zero and

to $-\infty$ at one, and is greater than a whenever $1+y+y^2<2$. Its maximum is therefore attained at some $y_*\in(0,1)$ and is greater than a. Moreover,

$$1+y+y^2-2\sqrt{y}=(1-\sqrt{y})^2+y^2>0,$$

so every value in (2), including its attained maximum, is less than $1/2$. At any maximizer y_*, equality holds in (3) for $\beta=\beta_c$. For every $\beta<\beta_c$, the same y_* gives a strict violation, with $z=y_*^3$ in (1).

A continuous family of signed densities

For $a<\beta<1/2$, define

$$H_\beta(s)=\frac{\Gamma(s+a)\Gamma(s+b)\Gamma(s+1)}{\Gamma(s+\beta)^2}, \tag{4}$$

$$w_{c,\beta}(u)=\frac{u^{c-1}(1-u)^{\beta-c}}{\Gamma(\beta+1-c)},\qquad c\in\{a,b\},\quad 0<u<1, \tag{5}$$

$$P_\beta(t)=t^2+(2\beta-3)t+(\beta-1)^2, \tag{6}$$

$$h_\beta(x)=\int_0^1\int_0^1 w_{a,\beta}(u)w_{b,\beta}(v)\frac{e^{-x/(uv)}}{uv}\,P_\beta\Big(\frac{x}{uv}\Big)\,du\,dv,\qquad x>0. \tag{7}$$

Both beta weights are integrable and positive. The function h_β may have either sign at this stage.

Lemma 2. *The family $h_\beta(x)$ is jointly continuous, with a continuous β derivative, on $(a,1/2)\times(0,\infty)$. For every complex s with $\Re s>-a$, it is absolutely integrable against x^s and satisfies*

$$\int_0^\infty x^s h_\beta(x)\,dx=H_\beta(s). \tag{8}$$

On each compact parameter interval $I\subset(a,1/2)$, it is positive for all sufficiently small $x>0$, uniformly in $\beta\in I$. It is also positive for every $x\ge 3$ and every $\beta\in(a,1/2)$.

Proof. For s in the stated half-plane, absolute Fubini is justified by putting $t=x/(uv)$ in (7). The absolute t integral is bounded by a sum of gamma integrals, since $\Re s>-a>-1$. The remaining beta integrals have exponents $s+a-1$ and $s+b-1$ at zero and integrable exponents at one. Furthermore,

$$\int_0^\infty t^s e^{-t}P_\beta(t)\,dt=(s+\beta)^2\Gamma(s+1),\qquad \int_0^1 u^s w_{c,\beta}(u)\,du=\frac{\Gamma(s+c)}{\Gamma(s+\beta+1)}.$$

Their product is (8). In particular $h_\beta\in L^1$. On compact positive x intervals, the exponential in (7) dominates all powers of $(uv)^{-1}$; logarithmic factors from a β derivative are integrable at $u=1,v=1$. This proves the asserted continuity by dominated differentiation.

Here are details of the uniform small-x assertion. Mellin inversion gives

$$h_\beta(x)=\frac{1}{2\pi i}\int_{-i\infty}^{i\infty}H_\beta(s)x^{-s-1}\,ds.$$

Indeed, (8) at $s=it$ is the Fourier transform of the integrable continuous function $e^v h_\beta(e^v)$, and is itself integrable. Uniform Stirling estimates on compact β intervals give

$$|H_\beta(\sigma+it)|\le C_I(1+|t|)^{A_I}e^{-\pi|t|/2}\quad(-1/2\le\sigma\le 0,\ |t|\ge 1).$$

We may therefore move the contour to $\Re s = -1/2$. Only the simple pole $s = -a$ is crossed. Consequently,

$$h_\beta(x) = C_\beta x^{-2/3} + R_\beta(x), \qquad C_\beta = \frac{\Gamma(1/3)\Gamma(2/3)}{\Gamma(\beta - 1/3)^2} > 0, \qquad |R_\beta(x)| \le M_I x^{-1/2}, \tag{9}$$

where one can take

$$M_I = \frac{1}{2\pi} \sup_{\beta \in I} \int_{\mathbb{R}} |H_\beta(-1/2 + it)|\, dt < \infty.$$

The minimum of C_β on I is positive, proving uniform positivity near zero. Finally, for $t \ge 3$ and $\beta > a$,

$$P_\beta(t) > t(t - 7/3) > 0.$$

When $x \ge 3$, the argument $t = x/(uv)$ in (7) is at least three. The entire integrand is then positive. □

Strict positivity at the threshold

Nonnegativity alone at the threshold would not suffice for perturbation on a compact x interval. The following argument establishes the needed pointwise strictness.

Lemma 3. *For every $x > 0$, $h_{\beta_c}(x) > 0$.*

Proof. The standard exponential formula for the gamma function yields, for $s \ge 0$,

$$\log \frac{H_\beta(s)}{H_\beta(0)} = d_\beta s + \int_0^\infty (e^{-su} - 1 + su) \frac{N_\beta(u)}{u}\, du, \tag{10}$$

where

$$d_\beta = \psi(a) + \psi(b) + \psi(1) - 2\psi(\beta), \qquad N_\beta(u) = \frac{e^{-au} + e^{-bu} + e^{-u} - 2e^{-\beta u}}{1 - e^{-u}}.$$

This is the same gamma/Lévy–Khintchine calculation used in [1, Section 2], with a change of scale. At $\beta = \beta_c$, (3) gives $N_\beta \ge 0$. Moreover $N_\beta(u)/u \sim u^{-2}$ at zero and decays exponentially at infinity. Thus (10) is the cumulant of an infinitely divisible real random variable T, with negative jumps and all positive exponential moments. The formula also holds on the imaginary axis. By (8) and uniqueness of Fourier transforms of finite signed measures, its continuous density is

$$g(t) = \frac{e^t h_{\beta_c}(e^t)}{H_{\beta_c}(0)}. \tag{11}$$

In particular, this expression is nonnegative everywhere.

Since $a < \beta_c < b < 1$, the term e^{-au} dominates the numerator of N_{β_c} at infinity. Choose $L > 0$ so that $N_{\beta_c}(u) > 0$ for all $u \ge L$. Choose a continuous bounded function ν_1 which vanishes for $u \le L$, is strictly positive for $u > L$, and satisfies

$$0 \le \nu_1(u) \le N_{\beta_c}(u)/u.$$

For example multiply that density on $u > L$ by $1 - e^{-(u-L)}$ and set it to zero elsewhere. Its mass $\Lambda = \int \nu_1(u)\, du$ is finite and positive, and ν_1 is uniformly continuous. Split the Lévy measure to write $T = W + Z$, independently, where Z is the negative compound-Poisson variable with intensity density ν_1. The remaining drift is absorbed in W.

For $s \geq 0$,

$$-\Lambda \leq \log \mathbb{E}e^{sZ} = \int_0^\infty (e^{-su} - 1)\nu_1(u)\, du \leq 0.$$

Stirling's formula gives $\log(H_{\beta_c}(s)/H_{\beta_c}(0)) = s\log s + O(s)$. It follows that $\log \mathbb{E}e^{sW} = s\log s + O(s)$, so W has unbounded support above: an upper bound $W \leq A$ would imply $\log \mathbb{E}e^{sW} \leq As$.

The event that Z has exactly one jump gives the continuous density minorant

$$g_1(t) = e^{-\Lambda} \int_{\mathbb{R}} \nu_1(w - t)\, \mathbb{P}(W \in dw).$$

For every real t, the unbounded upper support of W and the positivity of ν_1 on (L, ∞) imply $g_1(t) > 0$. Continuity follows from uniform continuity and boundedness of ν_1. The probability-measure domination gives $g \geq g_1$ almost everywhere; as both are continuous, it holds everywhere. Equation (11) now proves the lemma. □

Perturbation and the Mittag–Leffler transform

Choose a closed interval $I \subset (a, 1/2)$ with β_c in its interior. Let $c_I = \min_{\beta \in I} C_\beta > 0$ and choose $\eta \in (0, 1]$ so small that $M_I \eta^{1/6} < c_I/2$. By (9), $h_\beta(x) > 0$ for $0 < x \leq \eta$, uniformly for $\beta \in I$. Lemma 3 gives

$$m = \min_{\eta \leq x \leq 3} h_{\beta_c}(x) > 0.$$

Joint continuous differentiability from Lemma 2 gives

$$A = \sup_{\beta \in I,\ \eta \leq x \leq 3} |\partial_\beta h_\beta(x)| < \infty.$$

Take any positive ε smaller than both the distance from β_c to the endpoints of I and $m/[2(1 + A)]$. For $|\beta - \beta_c| < \varepsilon$, the mean value theorem gives $h_\beta(x) > m/2$ on $[\eta, 3]$. Together with the uniform small-x bound and positivity for $x \geq 3$, this proves

$$h_\beta(x) > 0 \quad \text{for all } x > 0, \qquad |\beta - \beta_c| < \varepsilon. \tag{12}$$

Normalize $h_\beta/H_\beta(0)$ to be the density of a positive random variable Y_β, and put $X_\beta = 3Y_\beta^{1/3}$. Gamma triplication and (8) give, for every $s \geq 0$,

$$\mathbb{E}X_\beta^s = 3^s \frac{H_\beta(s/3)}{H_\beta(0)} = \frac{\Gamma(1 + s)\Gamma(\beta)^2}{\Gamma(\beta + s/3)^2}. \tag{13}$$

The series for $F^{(2)}_{1/3,\beta}$ is entire, as follows directly from Stirling's formula. For every real $z \geq 0$, monotone convergence and (13) therefore give $\mathbb{E}e^{zX_\beta} = \Gamma(\beta)^2 F^{(2)}_{1/3,\beta}(z) < \infty$. Absolute convergence then justifies the same identity for complex z. In particular,

$$F^{(2)}_{1/3,\beta}(-x) = \frac{1}{\Gamma(\beta)^2} \mathbb{E}e^{-xX_\beta} \quad (x > 0). \tag{14}$$

For every integer $k \geq 0$, differentiation under this probability integral is justified by boundedness of $t^k e^{-xt/2}$ locally in $x > 0$. Since $X_\beta > 0$ almost surely, all alternating derivatives are strictly positive. Finally choose $\beta \in (\beta_c - \varepsilon, \beta_c)$. The threshold calculation shows failure of the proposed condition, while (14) proves complete monotonicity. This completes the proof of Theorem 1.

Task 2: Certify an explicit counterexample parameter

Replace the implicit positive width ε by a rigorous numerical lower bound. In particular, determine whether $\beta = 43/100$ is a counterexample, using certified bounds for the small-x remainder and the compact-interval minimum. A positive numerical grid alone is not a certificate.

Task 3: Determine the actual complete-monotonicity threshold

For $\alpha = 1/3$ and $\gamma = 2$, determine the infimum of the parameters β giving complete monotonicity. Describe the loss of positivity of h_β at this boundary, including whether it occurs through a unique double zero at a finite positive argument.

Task 4: Extend the counterexamples to other parameters

Establish an analogous stability theorem for other rational α and integer γ, and then for noninteger γ. Determine whether the counterexamples occupy a nonempty open subset of the full three-dimensional parameter space.

Task 5: Find a direct positive representation below the threshold

Construct an explicit positive mixture or product representation of X_β for counterexample parameters. Such a representation should make positivity visible without perturbing from β_c, and explain how the sign-changing gamma Lévy kernel is compatible with a positive probability law.

References

[1] T. Simon, *Remark on a Mittag–Leffler function of Le Roy type*, Integral Transforms Spec. Funct. **33** (2022), 108–114, doi:10.1080/10652469.2021.1913138. arXiv:2012.11018v2, 25 March 2021, Section 3, pages 5–6.